

EDUCATION

- **Rochester Institute of Technology (RIT)** Rochester, NY
Master of Science in Data Science; GPA: 3.9 Aug. 2023 – May 2026

EXPERIENCE

- **Morningstar** Mumbai, India
Associate Software Engineer May 2021 – May 2023
 - **End-to-End Product Ownership:** Built portfolio analytics tools with full ownership from data pipeline to user-facing interface in Vue.js, improving report clarity for 100k+ users.
 - **Scalable Content Platform:** Designed CMS-driven FAQ system using Contentstack APIs, enabling non-technical stakeholders to manage documentation independently.

RESEARCH

- **Dual-Objective Neural Architecture Search Ablation Study** Rochester, NY
Student ML Researcher, RIT Evolutionary Computing Lab Jan. 2025 – Present
 - **C++ Distributed Implementation:** Refactored C++ `SpeciationStrategy` classes to track evaluating genomes across MPI worker nodes, resolving async bottlenecks; orchestrated 50+ parallel ablation jobs on a Slurm-managed 2,304-core HPC cluster isolating effects of SWEET selection vs. Harada pruning on simultaneous MSE and size reduction.
 - **Statistical Validation:** Applied Mann-Whitney U tests ($p < 0.05$) across 4 time-series datasets to validate significance of dual-objective compression and accuracy gains.
- **Singh, A., Lyu, Z., Desell, T.** “Biologically-Inspired Homeostasis for Neuroevolution.” *EvoStar 2026* — 23% model compression ($p = 0.0019$) without MSE degradation on multivariate time-series forecasting.

PROJECTS

- **Generative AI Video Compliance & Brand Safety System**
LangGraph, Azure Video Indexer, Azure AI Search, LangSmith
 - **LLM-Based Content Quality:** Leveraged LLMs to detect policy violations and assess content quality across video and text modalities — directly applicable to ad quality scoring and brand safety enforcement at scale.
 - **LLMOps & Observability:** Integrated LangSmith tracing and Azure Monitor for prompt evaluation and latency monitoring, enabling iterative model improvement in production.
- **End-to-End Collaborative Filtering Recommendation System**
Python, Scikit-learn, Pandas, Streamlit, Docker
 - **Candidate Retrieval & Ranking:** Implemented user-based and item-based collaborative filtering via Nearest Neighbors for real-time recommendation retrieval and ranking analogous to ad candidate selection pipelines.
 - **ML Pipeline & Deployment:** Built modular ingestion-to-training pipeline; containerized with Docker and deployed Streamlit inference interface for end-to-end ownership from training to serving.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, JavaScript, Bash
- **ML & AI:** PyTorch, TensorFlow, Transformers (HuggingFace), Collaborative Filtering, Neural Architecture Search, Distributed Training (MPI/Slurm/HPC), XGBoost, Scikit-learn, RAG, LangChain, LangGraph, LLMOps
- **Infrastructure & Data:** FastAPI, Docker, Kubernetes, GCP, AWS, Azure (OpenAI, AI Search, Monitor), PostgreSQL, Pinecone, Pandas, NumPy, Airflow, Git